Eliguli Han

Academic records: Haocheng Han | New York, NY | hh3494@nyu.edu
Neuroscience portfolio | Full research portfolio | GitHub

Mathematics and computer science undergraduate seeking supervised research in computational neuroscience, sensory processing and multisensory perception. Background in scientific computing, volumetric image processing, acoustic simulation and statistical machine learning.

EDUCATION

New York University, College of Arts and Science

B.A. candidate, joint major in Mathematics and Computer Science | Anticipated graduation: 2028-2029

Selected completed coursework: Fundamentals of Machine Learning (A); Natural Language Processing (A-); Theory of Probability (A-); Partial Differential Equations; Numerical Analysis; Analysis; Linear Algebra and Differential Equations; Basic Algorithms. Current: Mathematical Statistics.

RESEARCH EXPERIENCE

Micro-CT Reconstruction and Computational Biomechanics

NYU DURF-supported research | Otodus megalodon tooth | 2026

- Developed a workflow for 3,050 Scanco micro-CT DICOM slices (>20 GB), including segmentation, marching-cubes surface extraction, Taubin smoothing, geometry repair and tetrahedral meshing; worked with a reconstruction of approximately 33 million triangles.
- Used an 18 micrometer z-step. Documented slice spacing separately from XY pixel spacing and avoided assuming isotropic voxels without calibration.
- Investigated finite-element material and geometric sensitivity in COMSOL; documented numerical verification, stress localization and interpretation limits in working technical reports.
- Transferable preparation for imaging research: data provenance, spatial calibration, geometric quality checks and sensitivity analysis.

Antiterrastrae: Biosonar and Acoustic Simulation

Self-directed computational prototype | In development

- Developing an aquatic simulation relating three-dimensional animal geometry to acoustic propagation, echoes and environmental parameters.
- Implemented a tetrahedral finite-element module for scalar acoustic eigenmodes, including stiffness and consistent mass matrices, a Cholesky transformation and Jacobi eigenvalue iteration.
- Integrated WebGL echo rendering with a Canvas fallback and VTK/OBJ geometry export for inspection and further analysis.

Research & selected work

Bone Architecture, Topology and Digital Cranial Anatomy

DURF / independent research development | Ongoing

- Developing finite-element and anatomically constrained topology-optimization approaches for fossil vertebral and cranial structures; exploring persistent homology, cubical complexes and discrete exterior calculus.
- Processing and presenting Cynthiacetus peruvianus cranial and mandibular geometry, with interactive 3D inspection and separate printable parts. Specimen reference: MNHN.F.PRU10.
- Exploring morphology, virtual endocast reconstruction and biomechanical questions as data availability permits.

Machine Learning, NLP and Information Retrieval

Coursework, collaborative software and independent implementation

- Implemented perceptron-based part-of-speech tagging with Viterbi decoding; obtained 96.9135% accuracy on the WSJ Section 24 evaluation split.
- Conducted retrieval experiments on approximately 315,000 records using TF-IDF, BM25 and RM3; developed search-service work using FastAPI, PostgreSQL/pgvector (HNSW) and Redis.
- Developed layered Ancient Greek semantic-graph analyses and mathematical probability / stochastic-process notes with inspectable computational examples.

PROPOSED NEUROSCIENCE DIRECTION

Vrontiki Synaisthesma - multisensory interaction research concept. Developing a long-term research and game-design framework for visual-haptic transfer, pseudo-haptic feedback and cross-species representations of sensory worlds. Seeking laboratory training to translate these questions into controlled behavioral tasks, quantitative models and, where relevant, neural-data analyses. Experimental and neural validation are proposed work.

SELECTED TECHNICAL WRITING & CODE

Working reports, course-project manuscripts and mathematical notes; publication status is stated separately for each item.

Tooth-scale Stress Localisation in a micro-CT-derived Otodus megalodon Model - working technical report

Compiling Ancient Greek Semantic DAGs with Phoneme, Morpheme, and Lexeme Layers - course-project manuscript

From Random Walks to Brownian Motion - mathematical notes

Selected implementation: fea_megalodon; Antiterrastrae; collaborative fml-project

METHODS, TOOLS & RESEARCH SUPPORT

Programming: Python, NumPy, SciPy, pandas, scikit-learn; JavaScript, HTML/CSS, WebGL/Canvas; SQL; Git.
Methods: numerical PDEs, finite-element assembly, eigenvalue problems, mesh processing, statistical evaluation, scientific visualization, information retrieval and sequence modeling.
Research tools: COMSOL, DICOM processing, TetWild, FastAPI, PostgreSQL/pgvector and Redis.
Support: NYU Dean's Undergraduate Research Fund (DURF), research project support.